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This section is a prerequisite. You must complete this section in order to unlock additional content.

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Đề thi chính thức

Final Exam due Aug 2, 2023 12:00 +07

Question #e2b1d2

1.0 point possible (graded, results hidden)

A neural network can be seen as a nonlinear function. That function is nonlinear in ...

- ☒ Both input signal x and weights
- ☐ Input signal x
- ☐ The weights

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Question #2cdfc5

1.0 point possible (graded, results hidden)

Which of the following can be the best initial cluster centroids selection for K-means ?

- ☒ Furthest cluster centroids
- ☐ None of the other options
- ☐ Closest cluster centroids
- ☐ Random cluster centroids

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Question #b0a09f

1.0 point possible (graded, results hidden)

Data used to optimize the parameter settings of a supervised model is called

- ☐ Training
- ☐ Test

✓ validation

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Question #d3e069

1.0 point possible (graded, results hidden)

In reinforcement learning, a policy is a function that maps ... to ...

- ☒ states to actions
- ☐ states to values
- ☐ values to actions
- ☐ actions to states
- ☐ values to states
- ☐ actions to values

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Question #43f9f3

1.0 point possible (graded, results hidden)

A multiple-layer neural network with linear activation functions is equivalent to one single-layer perceptron that uses the same error function on the output layer and has the same number of inputs.

- ☒ Yes
- ☐ No

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Question #5ddb52

1.0 point possible (graded, results hidden)

The perceptron algorithm will converge

- ☐ As long as you initialize weights to all 0's
- ☐ Always
- ☒ If the data is linearly separable
- ☐ Even if the data is linearly inseparable

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small gradient descent

0	x
x	0

can't

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- ☐ Choose a machine learning method for analyzing our datasets
- ☒ Transform (or fomulate) a real-life problem to be solved into a suitable machine learning problem
- ☒ Choose a machine learning model to solve our problem
- ☐ Transform (or fomulate) a real-life problem to be solved into an analytic problem

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Question #c3de93

1.0 point possible (graded, results hidden)

Denote $L(f, P)$ be the expected loss of binary classifier f on the whole data space, and $L(f, D)$ be the empirical loss of f on the training dataset D , where we use square loss to measure the error of f . What can happen if $|L(f, P) - L(f, D)| \leq 0.01$?

- ☐ f can be overfitting
- ☒ f can be underfitting ✓
- ☒ f can generalize well ✓
- ☐ f will perform extremely well on future data
- ☐ This case will never happen

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Question #95b9b1

1.0 point possible (graded, results hidden)

Which of the followings is not true about advantages of original K-means?

- ☐ K-means is ofen efficient and can converge quickly in practice.
- ☒ Different random initialization of k centroids will lead to the same resulting clusters.
- ☒ K-means can work with complex cluster shapes, such as non-convex data.

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Question #f610be

1.0 point possible (graded, results hidden)

Suppose you are using a Random Forest in model building. Which of the followings can be TRUE?

- 1. Number of trees should be as large as possible **F**
- 2. You can easily interpret why Random Forest make a prediction **F**

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☐ 2

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Question #a4c0ba

1.0 point possible (graded, results hidden)

Which of the following statements are true about linear regression?

- A. A linear regression model is learned by averaging the difference between the regression line and the training examples
- ☒ B. A linear regression model is often learned by minimizing the difference between the regression line and the training examples
- C. Linear regression can be used for both classification and regression problems
- ☒ D. Linear regression can be used for only classification problem
- ☒ E. Linear regression can be used for only regression problem

- ☐ A, B, C, D and E
- ☐ A and C
- ☐ B and C
- ☐ A and E
- ☒ B and E
- ☐ A, B and C

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Question #a4ddfb

1.0 point possible (graded, results hidden)

Consider the Gaussian mixture model (GMM) with density function

$$f(x) = \frac{1}{10}\mathcal{N}(x|1,1) + \frac{3}{10}\mathcal{N}(x|5,3) + \frac{4}{10}\mathcal{N}(x|8,2) + \frac{2}{10}\mathcal{N}(x|9,1)$$

where $\mathcal{N}(x|\mu, \sigma)$ is the density of the Gaussian distribution with mean μ and deviation σ .

Which Gaussian component may most likely generate example $x = 6$?

- ☐ the component with density $\mathcal{N}(x|5,3)$
- ☐ the component with density $\mathcal{N}(x|9,1)$
- ☐ the component with density $\mathcal{N}(x|1,1)$
- ☐ the component with density $\mathcal{N}(x|8,2)$

b, 0377

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- ☐ Normalizing the attributes of data
- ☐ Changing other distance measure
- ☐ Increase K to a very large value
- ☐ Weighting the attributes of data

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Question #403ac0

1.0 point possible (graded, results hidden)

When modeling data, uncertainty can appear in

- ☐ the collected data
- ☐ the learned parameter of the chosen model
- ☒ all the other options
- ☐ the model family we choose

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Question #f6c487

1.0 point possible (graded, results hidden)

When training a decision tree by ID3 algorithm, each attribute must be

- ☐ Continuous
- ☒ Nominal or categorical
- ☐ Of any form
- ☐ Numeric

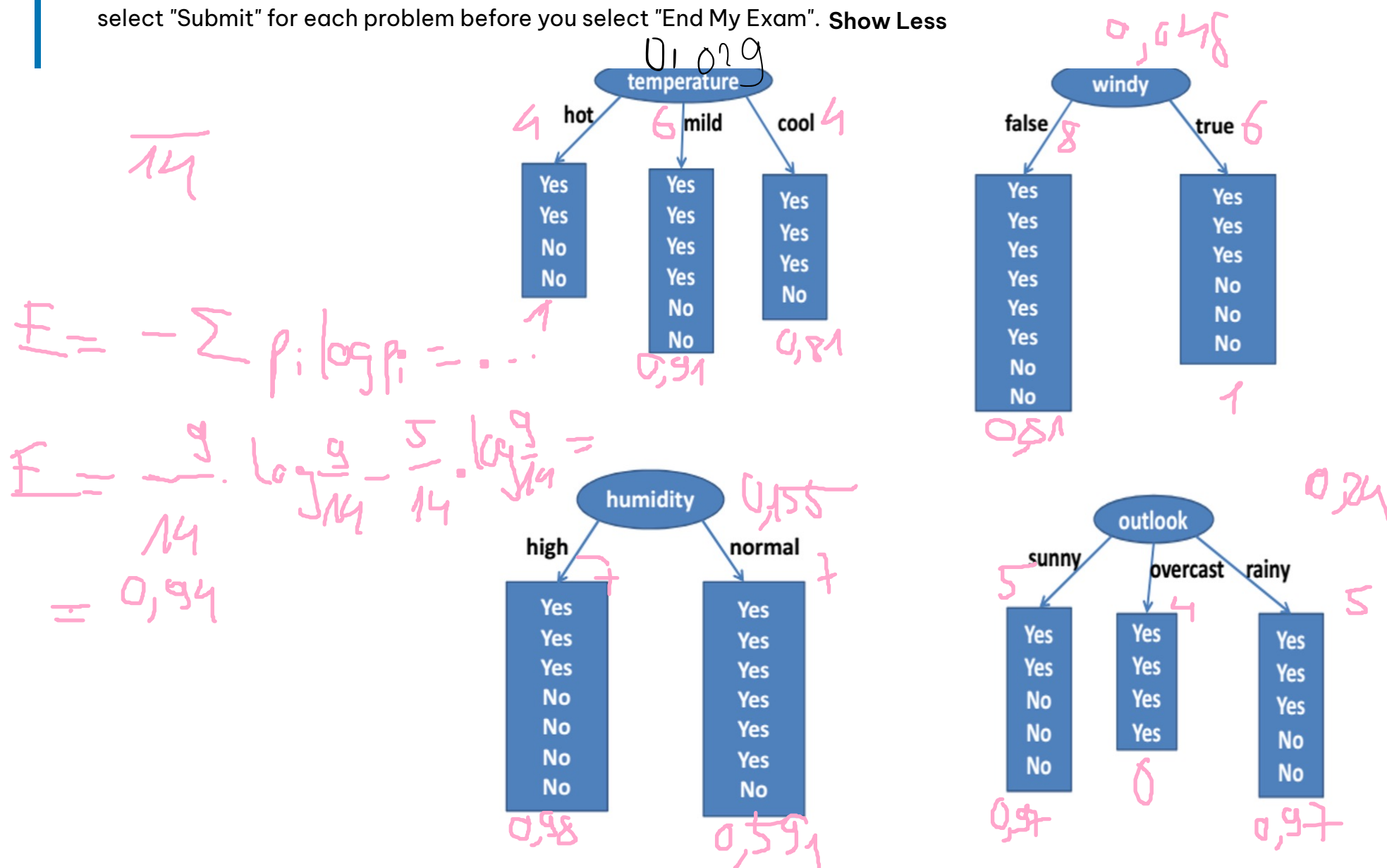
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Question #74ed31

1.0 point possible (graded, results hidden)

Suppose you are building decision tree model which split a node on the attribute, that has highest information gain. In the below image, select the attribute which has the highest information gain?

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- ☐ Temperature
- ☐ Humidity
- ☐ Windy
- ☒ Outlook

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Question #5ca820

1.0 point possible (graded, results hidden)

The complexity of each SVM training iteration depends on ...

- ☒ The size of the dataset
- ☒ The number of attributes
- ☐ The number of support vectors
- ☐ The penalty constant C

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- ☐ True
- ☐ False

?

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Question #396af2

1.0 point possible (graded, results hidden)

Which of the following are advantages of stacking (in Ensemble Learning)?

1. Better prediction.
2. More robust model.
3. Lower time of training and prediction.

- ☐ 1 and 3
- ☒ 1 and 2
- ☐ 1, 2 and 3
- ☐ 2 and 3

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Question #f1680a

1.0 point possible (graded, results hidden)

Consider a reinforcement learning agent in a non-deterministic world. Select the equation that correctly relates Q to value function v_π .

- ☒ $Q(s, a) = \sum_{s', r} P(s', r | a, s) [r + \gamma v_\pi(s')]$
- ☐ $Q(s, a) = \sum_{s', r} P(s', r | a, s) [r + v_\pi(s')]$
- ☐ $Q(s, a) = \sum_{s', r} P(s', r | a, s) \gamma [r + v_\pi(s')]$

✓

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Question #88c13c

1.0 point possible (graded, results hidden)

Which is INCORRECT in the following statements?

- ☐ Stratified sampling is suitable for small datasets. ✓
- ☐ Hold-out is suitable for large datasets. ✓
- ☐ Cross validation is suitable for small/average datasets. ✓

① ?

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Question #2c8c6a

1.0 point possible (graded, results hidden)

What are WRONG about Maximum A Posteriori estimation (MAP)?

- ☒ MAP needs not to know any knowledge about the parameters of a model $P(\theta)$
- ☐ MAP needs to know some knowledge about the parameters of a model
- ☒ MAP can estimate the full posterior distribution from a training dataset $\ln \theta$
- ☐ MAP can be used to learn a model from a training dataset
- ☐ MAP can make inference or prediction for new examples

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Question #4ba6a7

1.0 point possible (graded, results hidden)

In reinforcement learning, what is the reward hypothesis?

- ☒ Goals and purposes can be thought of as the maximization of the expected value of the cumulative sum of rewards received. ✓
- ☐ Always take the action that gives you the best reward at that point.
- ☐ Ignore rewards and find other signals.

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Question #91f7f0

1.0 point possible (graded, results hidden)

You are training a neural network with sigmoid activation functions. You discover that you are suffering from the vanishing gradient problem: with most of the training points, most of the components of the gradients are close to zero. It can cause training to be very slow. How could you combat with this problem?

- ☒ Make the network shallower (fewer layers). ✓
- ☒ Use ReLU activations instead of sigmoids. ✓
- ☐ Initialize the weights with larger values.
- ☐ Make the network deeper (more layers).

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- ☐ In case the number of the numerical attributes used for data representation is small, it performs better if they have the same scale.
- ☒ For any number of the numerical attributes used for data representation, it may perform better if they have the same scale. ✓
- ☐ In case the number of the numerical attributes used for data representation is large, it performs better if they have the same scale.

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Question #422862

1.0 point possible (graded, results hidden)

Consider Random Forest with K trees for regression. Each decision tree i can represent a function $T_i(x)$ of input x . Which of the following functions does Random Forest represent?

- ☒ $y(x) = \sum_{i=1}^K T_i(x) / K$ ✓
- ☐ $y(x) = \sum_{i=1}^K T_i(x/K)$
- ☐ $y(x) = \max_{i \in \{1, \dots, K\}} T_i(x)$
- ☐ $y(x) = \sum_{i=1}^K T_i(x)$

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Question #bd4b6d

1.0 point possible (graded, results hidden)

We want to train a linear model $f(x) = w_0 + w_1 x_1 + \dots + w_n x_n$ from a given training set of size M . In the case of $M \ll n$, what can happen with the LASSO method?

- ☒ There may be no unique solution
- ☒ There is a unique solution
- ☐ We cannot find any solution
- ☐ There is no solution

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Question #647e03

1.0 point possible (graded, results hidden)

Suppose you are working on a binary classification problem, and there are 3 models: M1 has 70% accuracy, M2 has 60% accuracy and M3 has 50% accuracy.

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- ☐ 100%
- ☐ 60%
- ☐ 50%

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Question #356ab0

1.0 point possible (graded, results hidden)

Assume that you train a classifier on 10,000 training points and obtain a training accuracy of 99%. However, when you submit it to Kaggle, your accuracy is only 67%. Which of the following has a good chance of improving your performance on Kaggle?

- ☐ Set your regularization coefficient (if any) to 0
- ☒ Train on more data
- ☒ Use a validation set to tune your hyperparameters
- ☐ Remove randomly some parts of the training set when training your classifier

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Question #16597

1.0 point possible (graded, results hidden)

In k-nearest neighbor method, a good initialization may improve much the overall performance.

- ☐ No, it's wrong
- ☐ Yes, of course
- ☒ Yes, with some suitable choices of similarity measure
- ☐ Yes, in some situations

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Question #d189

1.0 point possible (graded, results hidden)

What are WRONG in the following statements when training a linear SVM from a training set of size m ?

- ☒ The number of support vectors may depend on m

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Question #b82

1.0 point possible (graded, results hidden)

Suppose you have an unlabeled dataset: $\{x_1, x_2, \dots, x_N\}$. You run K-means with 50 different random initializations (always with the same value of k), and obtain 50 different clusterings of the data. What is the recommended way for choosing the good clustering?

- ☐ Always pick the final (50th) clustering, since by that time it is more likely to have converged to a good solution
- ☒ Choose the clustering with smallest $\frac{1}{N} \sum_{i=1}^N \|x_i - m_i\|^2$, where m_i is the mean of the cluster containing x_i ✓
- ☐ The only way to do so is if we also have labels y_i for our data

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Question #b7

1.0 point possible (graded, results hidden)

We want to train a linear model $f(x) = w_0 + w_1x$ from a given dataset $D = \{(1, -2); (1, -0.1); (3, 1.6); (-2.5, -2); (4.5, 0.1); (5, 3.3)\}$. After using Ordinary least squares (OLS), which of the followings is the correct statement for w_0 and w_1 ?

- ☐ $w_0 \in [0.4, 0.6]$ and $w_1 \in [0.6, 0.7]$
- ☐ $w_0 \in [-0.7, -0.5]$ and $w_1 \in [0.6, 0.8]$
- ☐ $w_0 \in [3, 4.5]$ and $w_1 \in [1, 2]$
- ☐ $w_0 \in [-1, 0]$ and $w_1 \in [1, 2]$
- ☐ $w_0 \in [-1.2, -0.8]$ and $w_1 \in [0.3, 0.6]$

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Question #3

1.0 point possible (graded, results hidden)

Which of the followings are TRUE about cross validation?

- ☐ Is guaranteed to prevent overfitting
- ☒ Is often used to select hyperparameters

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Question #88ba31

1.0 point possible (graded, results hidden)

What can happen when we use nonlinear SVM?

- ☐ Nothing
- ☒ A nonlinear SVM will be trained in a new feature space
- ☒ We are working in a space which is different from the input space
- ☒ A nonlinear classifier is used

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